

EE-482: Network Security and Cryptography

LECTURE NO 18: *Virus and Related Threats*

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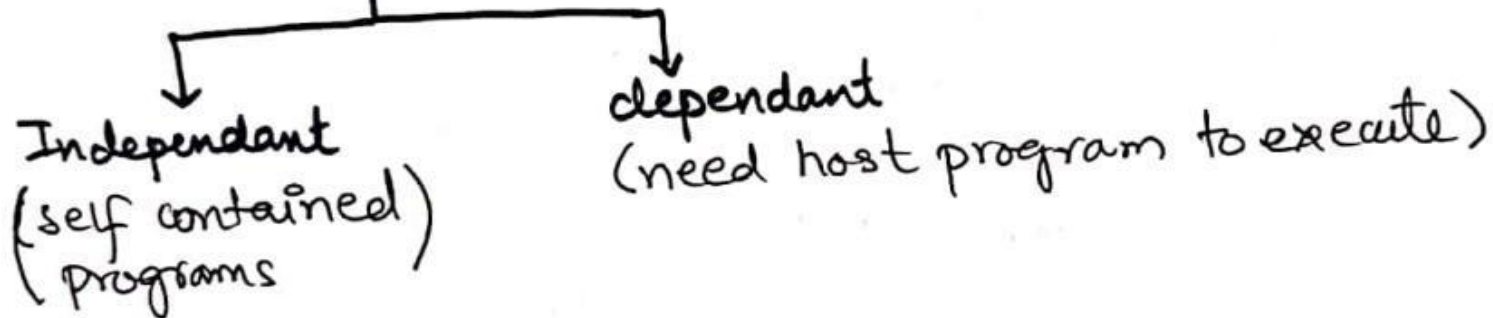
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20. Virus and Related Threats (1/11)

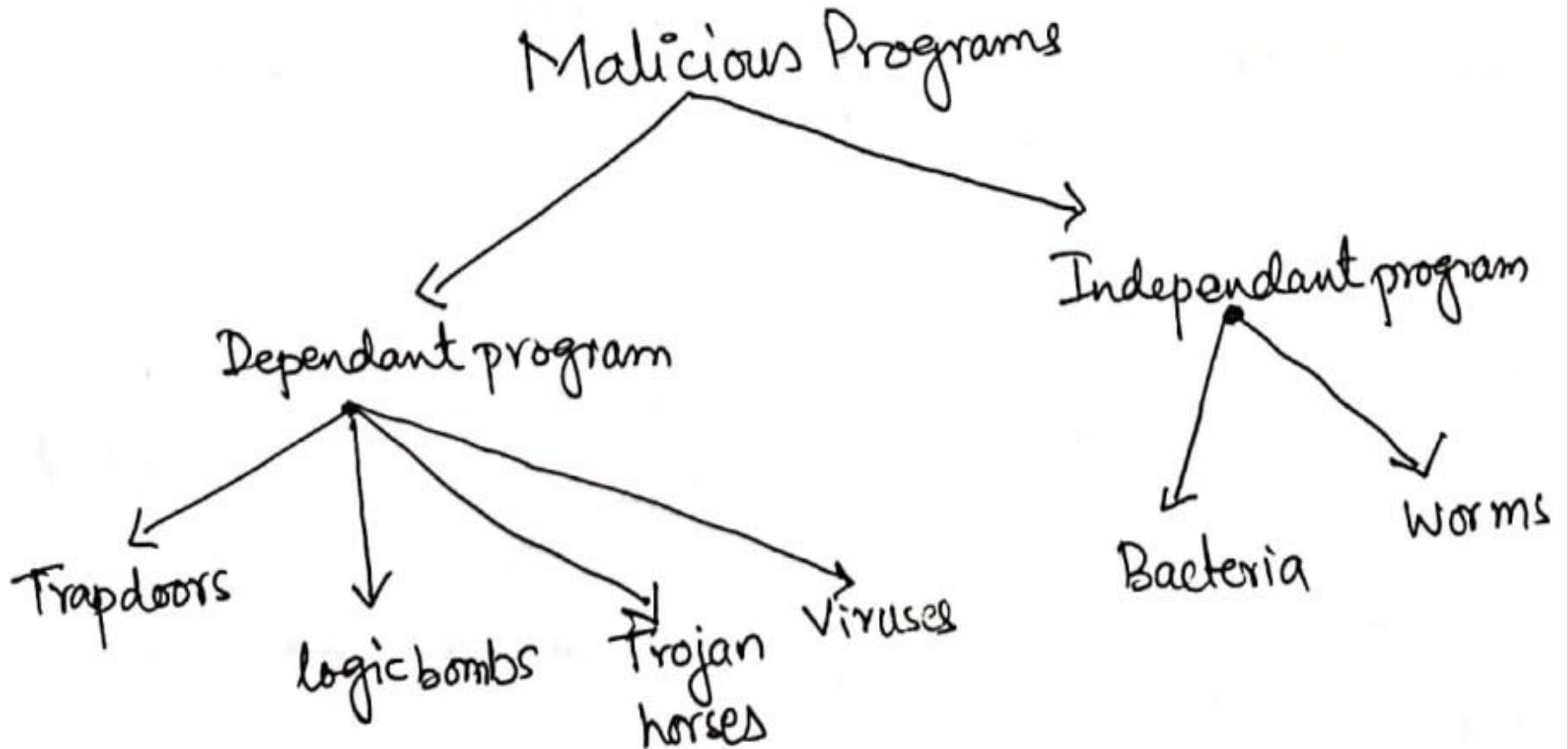
(a) Application Programs: For business & commercial application
e.g.: - inventory control programs
- payroll program

(b) Utility Programs: - Operating Systems
- Editors
- Compilers etc.

(c) Malicious Programs: - Undesirable/unwanted programs
- Threat to security
e.g.: virus, worms, trojan horses.



20. Virus and Related Threats (2/11)



20. Virus and Related Threats (3/11)

1) Trapdoors:

- ↳ A secret entry point into a program that allows someone that is aware of the trapdoor to gain access without through usual security access procedures.
- ↳ Trap door is a code that recognizes some special sequence of input or is triggered by being run from a certain user ID or by an unlikely sequence of events

20. Virus and Related Threats (4/11)

2) Logic Bombs:

↳ A code embedded in some legitimate program that is set to "explode" when certain conditions are met.

↳ Condition may be:

- presence/absence of certain files
- particular day of week or date
- particular user running the application

↳ Once triggered/exploded, a bomb may alter or delete data or entire file, cause a machine to halt etc.

20. Virus and Related Threats (5/11)

3) Trojan Horses

- ↳ A useful, or apparently useful program or command procedure containing a hidden/embedded code
- ↳ The code, when invoked, performs unwanted or harmful function.
- ↳ Trojan horse attempts to reveal confidential information to an attacker.

Example: A Trojan horse could silently sit in the code for a login screen by attaching itself to it.

- when user enters user id & password, the Trojan horse captures these details & sends to attacker without knowledge of user.

20. Virus and Related Threats (6/11)

4) Worms:

↳ Network worm programs use network connections to spread from system to system.

↳ Worm only replicates itself again & again.

↳ Doesn't perform any destructive action & only consumes system resources to bring it down.

↳ Replication grows so much that ultimately the system comes to a halt.

↳ To replicate itself a network worm uses some sort of network vehicle such as:

email facility: A worm mails a copy of itself to other systems.

Remote login capability: A worm logs onto a remote system as a user & then uses commands to copy itself from one system to other

20. Virus and Related Threats (7/11)

5) Bacteria:

- ↳ Programs that do not damage any files; instead their sole purpose is to replicate themselves.
- ↳ Bacteria reproduces exponentially, taking up all processor capacity, memory, disk space denying users access to these resources.

6) Zombie:

- ↳ it is a program that takes over another internet attached computer and then launches attack.

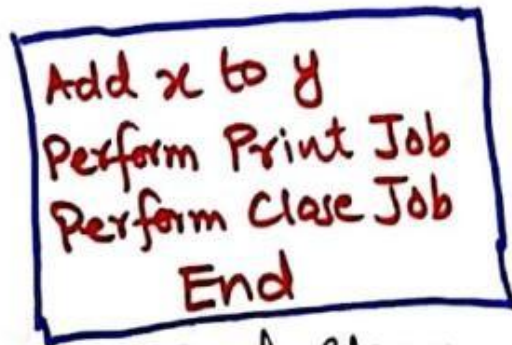
Examples: Malicious program will attack Alice's computer & send emails to all addresses in her address book.

- ↳ Can lead to increase in Network traffic & lead to Denial of Service Attack.

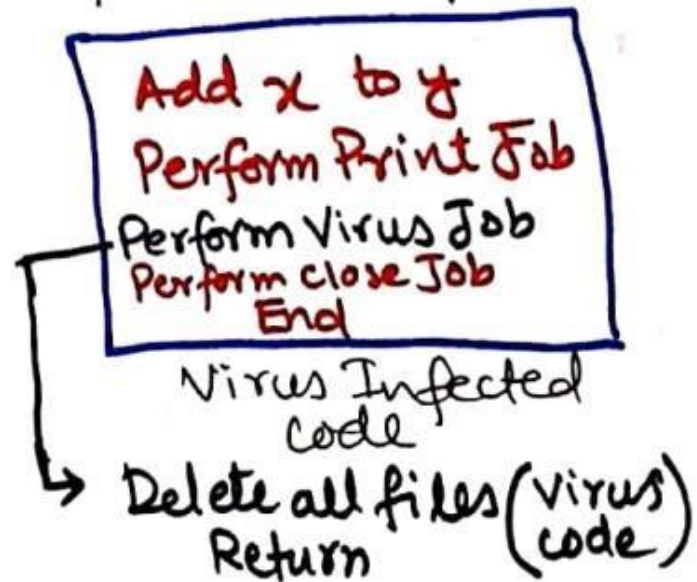
20. Virus and Related Threats (8/11)

7) Viruses:

- ↳ A program that can infect other programs by modifying them.
- ↳ Modification includes a copy of the virus program which goes on to infect other programs.
- ↳ Attaches itself to legitimate program code & runs when the legitimate program runs.
- ↳ Infection can spread from computer to computer



Original Clean
code



Virus Infected
code

20. Virus and Related Threats (9/11)

↳ A virus can be repaired and its damage can be controlled by using good backup procedures.

Lifetime of Viruses:

Divided into 4 phases

- (1) Dormant phase
- (2) Propagation Phase
- (3) Triggering Phase
- (4) Execution Phase

(1) Dormant Phase:

- ↳ Virus is idle, not doing anything
- ↳ The virus will be activated by some event such as
- a specific date
 - presence of another program
 - capacity of disk exceeding some limit

20. Virus and Related Threats (10/11)

(2) Propagation Phase

- ↳ virus places an identical copy of itself into another program
- ↳ each infected program will contain a clone of the virus which will itself enter a propagation phase.

20. Virus and Related Threats (11/11)

(3) Triggering Phase:

- ↳ Virus is activated to perform the function for which it was intended.
- ↳ This activation is to start destruction as opposed to activation in dormant phase where activation is to start propagation.
- ↳ Can be caused by happening of an event such as the number of copies of virus.

(4) Execution Phase:

- ↳ Actual function is performed for which virus is created
- ↳ Function may be harmless or damaging (destruction of programs & data files).

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*